# gpeyt1-39 · Uniform surcharge thrust

GEO PE · Long Calculation Problems I — Topic 8: Retaining Structures

## Problem

A 16 ft basement-type wall that can yield to the active state retains dry, level silty sand (γ = 115 pcf, φ = 30°, c = 0). A uniform surcharge of q = 400 psf from a parking area acts over the full backfill. No water; ignore wall friction. The designer needs the total thrust and its height above the base for the moment check.

**Most nearly, the total thrust and its height above the base are:**

A. 7,040 lb/ft at 6.1 ft
B. 7,040 lb/ft at 5.3 ft
C. 11,307 lb/ft at 6.8 ft
D. 5,973 lb/ft at 5.3 ft

## Given equations / data

> $\sigma h = K_a(q + \gamma z)$; $P_q = K_a q H$ at $H/2$; $P_s = \frac{1}{2}K_a\gamma H^2$ at $H/3$

## Worked solution

1. $K_a = 1/3$
2. $P_q = (1/3)(400)(16) = 2{,}133$ lb/ft at 8 ft
3. $P_s = \frac{1}{2}(1/3)(115)(16)^2 = 4{,}907$ lb/ft at 5.33 ft
4. $P = 7{,}040$ lb/ft; $\bar{y} = (2{,}133\times8 + 4{,}907\times5.33)/7{,}040 = 6.14$ ft

## Final answer

**7,040 lb/ft at 6.1 ft**

## Why the other values are wrong

- **7,040 lb/ft at 5.3 ft** — placed the whole resultant at H/3
- **11,307 lb/ft at 6.8 ft** — applied full q (forgot Ka on the surcharge)
- **5,973 lb/ft at 5.3 ft** — treated the surcharge as triangular

## References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.2 Rankine; §3.13.5.1 σhq = Ka q; §3.1.4 Load Distribution from Surcharge

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=4cCtjNpcFkA (no creator values copied).

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